

Full Stack Development with MERN

Project Documentation format

1. Introduction

- **Project Title:** Complaint ease: Online Registrations And Management Systems
- **Team Members:** Individual Project

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2. Project Overview

Purpose

Complaint Ease is a web-based complaint registration and management system developed using the MERN stack. The system allows users to submit complaints online, track complaint progress, and communicate with assigned agents. Administrators can manage complaints, users, and agents through a centralized dashboard.

The project aims to improve transparency, reduce manual complaint handling, and provide efficient complaint resolution.

Features

- User Registration & Login
- JWT Authentication
- Role-Based Access (User, Agent, Admin)
- Create Complaint
- Complaint Status Tracking
- Complaint Assignment
- Admin Dashboard
- Agent Dashboard
- Complaint History
- Feedback System
- Responsive User Interface

3. Architecture

Frontend Architecture (React.js)

The frontend is developed using React.js with component-based architecture. Tailwind CSS is used for responsive UI design.

Frontend Technologies

- React.js
- Tailwind CSS
- React Router DOM
- Axios
- Context API

Frontend Modules

- Authentication Pages
- User Dashboard
- Agent Dashboard
- Admin Dashboard
- Complaint Tracking
- Chat Window
- Complaint Management

Backend Architecture (Node.js & Express.js)

The backend follows MVC architecture.

Backend Technologies

- Node.js
- Express.js
- JWT Authentication
- bcryptjs
- express-validator
- dotenv
- cors
- multer

Backend Modules

- Authentication Controller
- Complaint Controller
- Admin Controller
- Agent Controller
- Feedback Controller

Database Architecture (MongoDB)

MongoDB is used as the NoSQL database.

Collections

Users

```
{  
  "name": "string",  
  "email": "string",  
  "password": "string",  
  "role": "USER | AGENT | ADMIN"  
}
```

Complaints

```
{  
  "title": "string",  
  "description": "string",  
  "category": "string",  
  "status": "Pending | Assigned | In Progress | Resolved",  
  "userId": "ObjectId",  
  "agentId": "ObjectId"  
}
```

Feedback

```
{  
  "rating": "number",  
  "comment": "string",  
  "complaintId": "ObjectId"  
}
```

4. Setup Instructions

Prerequisites

Install the following software:

- Node.js
- MongoDB
- VS Code
- Git

Installation Steps

Clone Repository

```
git clone:https://github.com/srisoumya06/VIP-C2-Complaint-Ease-Online-Registrations-and-Management-System.git
```

Frontend Setup

```
cd frontend  
npm install  
npm start
```

Backend Setup

```
cd Backend  
npm install  
npm start
```

```
Environment Variables (.env)
PORT=5000
MONGO_URI=your_mongodb_connection
JWT_SECRET=your_secret_key
```

5. Folder Structure

Frontend Structure (client structure)

```
Frontend/
|
|— src/
|   |— components/
|   |   |— admin/
|   |   |— agent/
|   |   |— common/
|   |   |— user/
|   |
|   |— Images/
|   |— App.js
|   |— index.js
|   |— App.css
```

Backend Structure(server structure)

```
Backend/
|
|— config/
|— controllers/
```

|— middleware/

|— models/

|— routes/

|— uploads/

|— server.js

|— .env

6. Running the Application

Frontend

```
cd frontend  
npm start
```

Frontend runs on:

`http://localhost:3000`

backend

```
cd backend  
npm start
```

Backend runs on:

`https://localhost:5000`

7. API Documentation

Authentication APIs

Register User

`POST /api/auth/register`

Login User

`POST /api/auth/login`

Complaint APIs

Create Complaint

`POST /api/complaints`

Get Complaints

`GET /api/complaints`

Update Complaint

`PUT /api/complaints/:id`

Delete Complaint

DELETE /api/complaints/:id

Admin APIs

Assign Complaint

PUT /api/admin/assign-agent/:id

View All Users

GET /api/admin/users

8.Authentication

Authentication is implemented using JWT (JSON Web Token).

Authentication Flow

User Login

↓

JWT Token Generated

↓

Token Stored in Local Storage

↓

Protected Routes Accessed

Security Features

- Password Hashing using bcryptjs
- JWT Token Verification
- Protected APIs
- Role-Based Authorization

9.User Interface

UI Modules

- Home Page
- Login Page
- Signup Page
- User Dashboard
- Admin Dashboard
- Agent Dashboard
- Complaint Status Page
- Chat Window

UI Design

- Responsive Layout

- Tailwind CSS Styling
- Dashboard Cards
- Sidebar Navigation
- Toast Notifications

10. Testing

Testing Strategy

- API Testing using Postman
- Frontend Testing using Browser Dev Tools
- MongoDB Database Testing

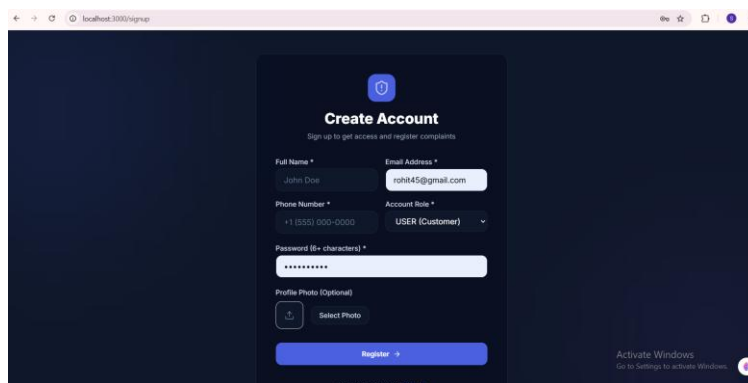
Tested Modules

- Authentication
- Complaint CRUD
- Status Updates
- Admin Assignment

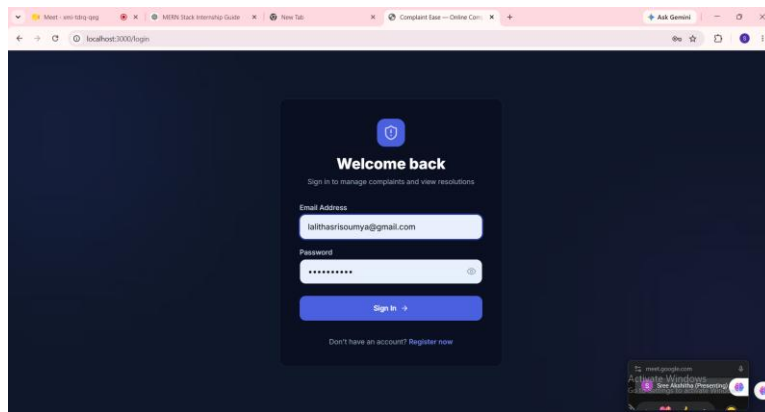
11. Screenshots or Demo

Screenshots

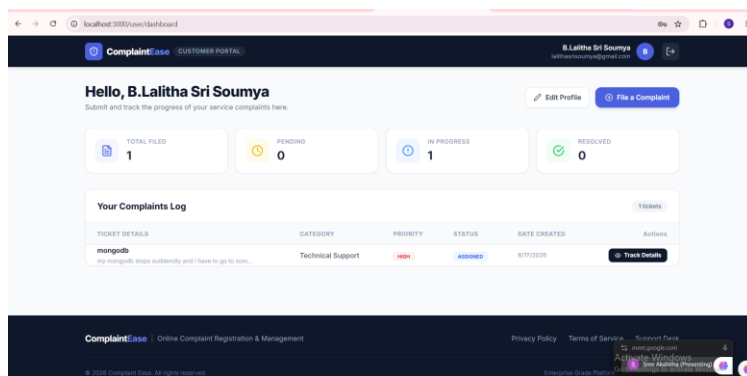
Signup page:



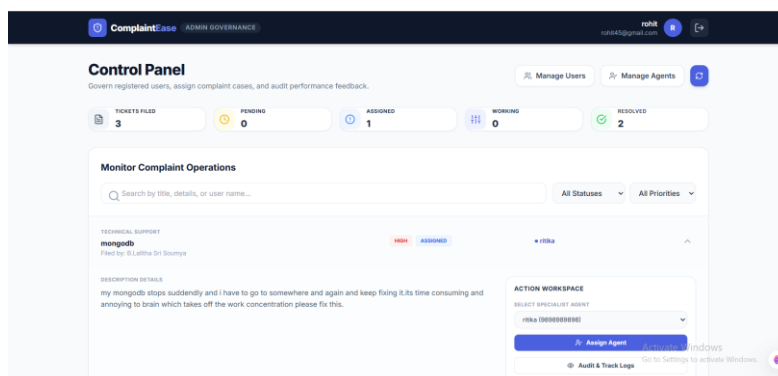
Login page:



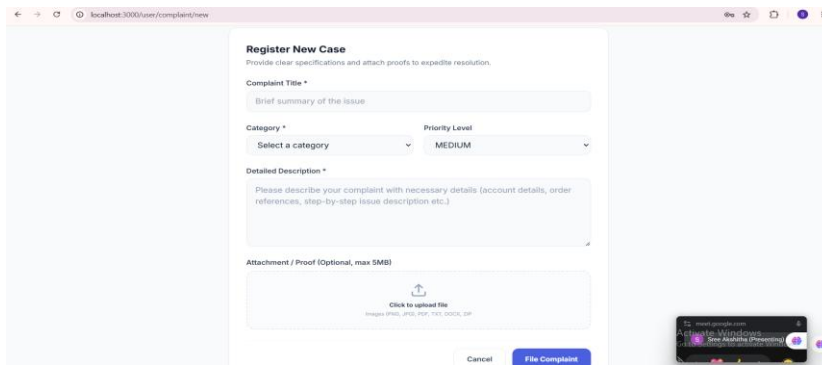
User dashboard:



Admin Dashboard:



Complaint status page:



The screenshot shows a web browser window with the address bar displaying 'localhost:3000/user/complaint/new'. The page title is 'Register New Case'. Below the title, there is a subtitle 'Provide clear specifications and attach proofs to expedite resolution.' The form contains the following fields:

- Complaint Title ***: A text input field with the placeholder 'Brief summary of the issue'.
- Category ***: A dropdown menu with the placeholder 'Select a category'.
- Priority Level**: A dropdown menu with the value 'MEDIUM' selected.
- Detailed Description ***: A text area with the placeholder 'Please describe your complaint with necessary details (account details, order references, step-by-step issue description etc.)'.
- Attachment / Proof (Optional, max 5MB)**: A file upload area with a 'Click to upload file' button and a list of supported file types: 'Images (PNG, JPG, GIF, PDF, DOCX, ZIP)'.

At the bottom of the form, there are two buttons: 'Cancel' and 'File Complaint'.

Include:

- Signup Page
- Login Page
- User Dashboard
- Admin Dashboard
- Complaint Status Page

Demo

GitHub Repository Link: <https://github.com/srisoumya06/VIP-C2-Complaint-Ease-Online-Registrations-and-Management-System.git>

12. Known Issues

- Real-time notifications not implemented
- Chat module under development
- File upload optimization pending

13. Future Enhancements

- Real-Time Chat using Socket.io
- Email Notifications
- SMS Alerts
- AI-Based Complaint Categorization
- Analytics Dashboard
- Mobile Application Support